Joint Tracking of Correlated Quantities in Cert-RNN+: Mathematical Formulation

Cert-RNN+ benchmark notes

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Abstract

Cert-RNN+ certifies that an LSTM-autoencoder anomaly detector raises no false alarm for any input in a perturbation box. It is built on the zonotope (affine-arithmetic) abstract domain, in which every intermediate quantity is an affine function of a shared vector of noise symbols. This note states the soundness invariant of that domain and derives the two refinements that distinguish Cert-RNN+ from Cert-RNN: (i) *linked gates*, which bound each bilinear gate product over the joint reachable region of its two operands instead of over their bounding box, and (ii) a *joint score bound*, which bounds the reconstruction error of all outputs together instead of component by component. Both refinements are sound and never looser than the baseline. We also explain why box-based linear relaxations, as used by CROWN, cannot exploit the same structure.

1 Setting

Detector. An LSTM autoencoder maps a window $x \in \mathbb{R}^{T\times D}$ (T seconds, D features) to a reconstruction $\hat{x} = \mathrm{AE}(x)$. The encoder LSTM produces one code per second; the decoder LSTM reads the code of second t at step t, followed by a linear head. The anomaly score is the mean squared reconstruction error

$$s(x) \;=\; \frac{1}{N}\,\|\mathrm{AE}(x) - x\|_2^2\,, \qquad N = TD, \tag{1}$$

and an alarm is raised iff $s(x) > \tau$, where τ is a threshold fixed from validation data.

LSTM step. With pre-activations $a^{(\cdot)} = W^{(\cdot)}x_t + U^{(\cdot)}h_{t-1} + b^{(\cdot)}$, one step is

$$i_t = \sigma(a^{(i)}),\ f_t = \sigma(a^{(f)}),\ g_t = \tanh(a^{(g)}),\ o_t = \sigma(a^{(o)}),\quad c_t = f_t\odot c_{t-1} + i_t\odot g_t,\quad h_t = o_t\odot\tanh(c_t). \tag{2}$$

The non-linear operations are therefore of two kinds: unary activations (σ, tanh) and *bilinear products of correlated operands*: $\sigma(\cdot)\cdot\tanh(\cdot)$ (in $i_t\odot g_t$ and $o_t\odot\tanh(c_t)$) and $\sigma(\cdot)\cdot(\cdot)$ (in $f_t\odot c_{t-1}$).

Property (no false alarm). Let x^0 be a normal window with $s(x^0) \le \tau$, let $F \subseteq \{1,\dots,T\}$ be the attacked seconds, and let $\epsilon > 0$. The perturbation set is

$$X_\epsilon = \big\{\, x : \big|x_{t,d} - x^0_{t,d}\big| \le \epsilon \text{ for } t \in F,\ x_{t,d} = x^0_{t,d} \text{ for } t \notin F \,\big\}. \tag{3}$$

The property is $\forall x \in X_\epsilon :\ s(x) \le \tau$. A verifier proves it by computing a sound upper bound $\bar{s} \ge \max_{x\in X_\epsilon} s(x)$ and checking $\bar{s} \le \tau$.

2 Zonotopes and affine forms

Definition 1 (Affine form, zonotope)**.** Fix a vector of noise symbols $\varepsilon \in [-1,1]^p$. An *affine form* is $\hat{z} = c + g^\top \varepsilon$ with $c \in \mathbb{R}$, $g \in \mathbb{R}^p$. A vector of n affine forms sharing the same symbols, $\hat{z} = c + G\varepsilon$ with $G \in \mathbb{R}^{n \times p}$, denotes the *zonotope*

$$Z(c, G) = \{\, c + G\varepsilon : \varepsilon \in [-1,1]^p \,\} \subseteq \mathbb{R}^n. \tag{4}$$

Geometrically, $Z(c,G)$ is the Minkowski sum of the p segments $[-G_{:,j}, G_{:,j}]$, translated by c; it is a centrally symmetric convex polytope [8]. The interval hull (bounding box) of coordinate k is $[c_k - \|G_{k,:}\|_1\,,\ c_k + \|G_{k,:}\|_1]$.

Input encoding. Assign one symbol to every attacked reading. Then $X_\epsilon = Z(x^0, G_0)$ exactly, where G_0 has entry ϵ in the row of reading (t,d) and the column of its symbol, and zeros elsewhere. Every quantity computed from the input is subsequently expressed over the *same* symbols, plus fresh symbols introduced by non-linear steps.

The soundness invariant. Abstract interpretation [1] with affine arithmetic [2, 3] maintains:

> **(I)** For every concrete input $x \in X_\epsilon$ there exists an assignment $\varepsilon \in [-1,1]^p$ of *all* symbols (input and fresh) such that every tracked quantity equals the value of its affine form at ε.

Invariant (I) is what allows several quantities to be reasoned about *jointly*: a single assignment ε is one concrete scenario, and every affine form is evaluated in that same scenario. Two forms that share symbols are therefore correlated, and the correlation is exact. Zonotope analyses of neural networks build on this [4, 5, 6], including Cert-RNN for recurrent networks [7].

Proposition 2 (Affine operations are exact)**.** *If $\hat{z} = c + G\varepsilon$ satisfies (I), so does $W\hat{z} + b = (Wc + b) + (WG)\varepsilon$, with no approximation.*

Proof. For the assignment ε given by (I), $Wz + b = W(c + G\varepsilon) + b$. □

Proposition 3 (Unary non-linearities)**.** *Let $\hat{z} = c + g^\top \varepsilon$ satisfy (I), with interval hull $[l,u]$, and let $\phi \in \{\sigma, \tanh\}$. Choose $a, b \in \mathbb{R}$ and $\beta \geq \max_{z \in [l,u]} |\phi(z) - (az + b)|$. Then*

$$\widehat{\phi(z)} = a\hat{z} + b + \beta\, \varepsilon_{\text{new}} \tag{5}$$

with a fresh symbol $\varepsilon_{\text{new}} \in [-1,1]$ satisfies (I).

Proof. For the concrete value $z \in [l,u]$, set $\varepsilon_{\text{new}} = (\phi(z) - az - b)/\beta \in [-1,1]$ (any value if $\beta = 0$). □

Remark 4. The term $a\hat{z}$ keeps all symbols of $\hat{z}$, so the output stays correlated with every other quantity that shares them; only the error part $\beta\varepsilon_{\text{new}}$ is independent. Cert-RNN's specific choices of (a, b, β) [7] are used unchanged; the refinements below concern only bilinear products and the score.

3 The joint reachable region of two correlated quantities

Consider two tracked quantities that share symbols,

$$\begin{pmatrix}\hat{x}\\ \hat{y}\end{pmatrix} = \begin{pmatrix}c_x\\ c_y\end{pmatrix} + \begin{pmatrix}g_x^\top\\ g_y^\top\end{pmatrix}\varepsilon, \qquad \varepsilon \in [-1,1]^p. \tag{6}$$

Lemma 5 (Joint region is a zonogon)**.** *The set of pairs reachable under (I), $Z_{xy} = \{(c_x + g_x^\top\varepsilon,\ c_y + g_y^\top\varepsilon) : \varepsilon \in [-1,1]^p\}$, is the image of the cube $[-1,1]^p$ under a linear map $\mathbb{R}^p \to \mathbb{R}^2$: a centrally symmetric convex polygon with at most $2p$ edges (a* zonogon*). It satisfies*

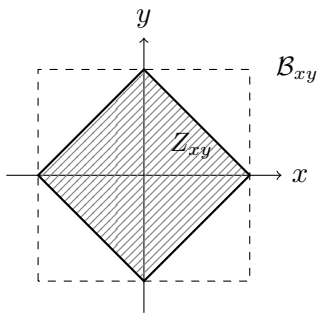
$$Z_{xy} \subseteq \mathcal{B}_{xy} \coloneqq [c_x \pm \|g_x\|_1] \times [c_y \pm \|g_y\|_1], \tag{7}$$

and a corner of $\mathcal{B}_{xy}$ belongs to Z_{xy} only if a single ε attains both coordinate extremes, i.e. only if $\operatorname{sign}(g_{x,j})$ *and* $\operatorname{sign}(g_{y,j})$ *agree (or disagree) consistently over all j with $g_{x,j}g_{y,j} \neq 0$. Its area is*

$$\operatorname{area}(Z_{xy}) = 4 \sum_{1\le i<j\le p} \left|\det \begin{pmatrix} g_{x,i} & g_{x,j}\\ g_{y,i} & g_{y,j}\end{pmatrix}\right|. \tag{8}$$

Proof. The image of a cube under a linear map is a zonotope; in $\mathbb{R}^2$ it is a polygon whose edges are the generator vectors $(g_{x,j}, g_{y,j})$, each used twice, sorted by angle, which gives at most $2p$ edges and central symmetry. The upper-right corner $(c_x + \|g_x\|_1, c_y + \|g_y\|_1)$ requires $\varepsilon_j = \operatorname{sign}(g_{x,j})$ and $\varepsilon_j = \operatorname{sign}(g_{y,j})$ for every j, which is possible only if the signs agree wherever both are non-zero; the other corners are analogous. Formula (8) is the area of a Minkowski sum of the segments $[-v_j, v_j]$ with $v_j = (g_{x,j}, g_{y,j})$: each pair of segments contributes a parallelogram of area $|\det(2v_i, 2v_j)| = 4\,|\det(v_i, v_j)|$. □

Example 6. Let $\hat{x} = \varepsilon_1 + \varepsilon_2$ and $\hat{y} = \varepsilon_1 - \varepsilon_2$. Each ranges over $[-2,2]$, so $\mathcal{B}_{xy} = [-2,2]^2$ with area 16. But $x = 2$ forces $\varepsilon_1 = \varepsilon_2 = 1$ and hence $y = 0$. The joint region is the diamond with vertices $(\pm 2, 0)$ and $(0, \pm 2)$, of area 8 by (8); none of the four box corners is reachable.



In an LSTM, the two operands of every gate product are computed from the same inputs and the same previous state, so they share many symbols and Z_{xy} is typically a thin slanted polygon well inside $\mathcal{B}_{xy}$.

4 Linked gates: bilinear products over the joint region

Let $f(x,y)$ be one of the bilinear products in (2), $f = \sigma(x)\tanh(y)$ or $f = \sigma(x)\,y$, applied coordinate-wise to operands $(\hat{x}, \hat{y})$.

Definition 7 (Plane relaxation over a region)**.** For slopes $A, B \in \mathbb{R}$ (chosen as in Cert-RNN [7]) and a set $S \subseteq \mathbb{R}^2$, let

$$C_1(S) = \inf_{(x,y)\in S} \big(f(x,y) - Ax - By\big), \qquad C_2(S) = \sup_{(x,y)\in S} \big(f(x,y) - Ax - By\big). \tag{9}$$

Proposition 8 (Soundness of a plane relaxation)**.** *If $S \supseteq Z_{xy}$, then*

$$\widehat{f(x,y)} = A\hat{x} + B\hat{y} + \frac{C_1(S) + C_2(S)}{2} + \frac{C_2(S) - C_1(S)}{2}\,\varepsilon_{\text{new}} \tag{10}$$

with a fresh symbol $\varepsilon_{\text{new}} \in [-1, 1]$ satisfies (I).

Proof. Under (I), the concrete pair (x, y) lies in $Z_{xy} \subseteq S$, so $C_1(S) \le f(x,y) - Ax - By \le C_2(S)$; choose ε_{new} accordingly. □

The width $C_2(S) - C_1(S)$ is the new, uncorrelated error that enters the cell state c_t and is carried to all later steps. Cert-RNN uses $S = \mathcal{B}_{xy}$. Cert-RNN+ uses a set between Z_{xy} and $\mathcal{B}_{xy}$:

Definition 9 (Grid cover)**.** Partition $\mathcal{B}_{xy}$ into an $n \times n$ grid of axis-aligned cells $Q_1, \ldots, Q_{n^2}$ and keep the cells that intersect the zonogon: $\mathcal{A} = \{Q_k : Q_k \cap Z_{xy} \neq \emptyset\}$. Set $S_+ = \bigcup_{Q\in\mathcal{A}} Q$. Then

$$C_1(S_+) = \min_{Q\in\mathcal{A}} C_1(Q), \qquad C_2(S_+) = \max_{Q\in\mathcal{A}} C_2(Q), \tag{11}$$

where each $C_i(Q)$ is the exact extremum of $f - Ax - By$ over a box, computed by the same closed-form box enumeration Cert-RNN uses for $\mathcal{B}_{xy}$.

Theorem 10 (Linked gates are sound and never looser)**.**

$$Z_{xy} \;\subseteq\; S_+ \;\subseteq\; \mathcal{B}_{xy}, \tag{12}$$

hence (10) *with $S = S_+$ satisfies (I), and*

$$C_1(S_+) \ge C_1(\mathcal{B}_{xy}), \qquad C_2(S_+) \le C_2(\mathcal{B}_{xy}), \qquad C_2(S_+) - C_1(S_+) \le C_2(\mathcal{B}_{xy}) - C_1(\mathcal{B}_{xy}). \tag{13}$$

Proof. Every point of $Z_{xy} \subseteq \mathcal{B}_{xy}$ lies in some cell, and that cell intersects Z_{xy}, so it is in $\mathcal{A}$; hence $Z_{xy} \subseteq S_+$. Each cell is a subset of $\mathcal{B}_{xy}$, hence $S_+ \subseteq \mathcal{B}_{xy}$. Soundness follows from the previous proposition. An infimum over a subset is at least the infimum over the superset, and a supremum over a subset is at most the supremum over the superset. □

Deciding cell admission exactly. A cell Q (a rectangle) and Z_{xy} (a convex polygon) are disjoint iff some axis separates them; by the separating-axis theorem for convex polygons it suffices to test the two coordinate axes and the edge normals of Z_{xy}, i.e. the normals of the generator vectors [9, 10]. The support of Z_{xy} along a direction w is $w^\top(c_x, c_y) \pm \sum_j \left|w^\top(g_{x,j}, g_{y,j})\right|$, computed exactly from all generators; the implementation adds a small outward tolerance, which can only admit more cells and therefore preserves soundness.

Two shortcuts that preserve soundness.

1. *Area-ratio skip.* If $\text{area}(Z_{xy})/\,\text{area}(\mathcal{B}_{xy}) > 0.9$, computed in closed form by (8) before any cell is built, the zonogon nearly fills its box and the cover can recover little; the box result $S = \mathcal{B}_{xy}$ is used. This is sound because the box result is always sound, and it trades only tightness for time.

2. *All-cells equality.* If every cell is admitted, $\mathcal{A}$ is the whole grid and $S_+ = \mathcal{B}_{xy}$; then $C_i(S_+) = C_i(\mathcal{B}_{xy})$ exactly, so the box result is returned without loss.

Grid resolution. The implementation uses $n = 16$ per axis. It was chosen in a development measurement (on models separate from the benchmark) of the fraction of the exact box-to-zonogon improvement recovered: 0.62–0.69 at $n = 8$, 0.80–0.85 at $n = 16$ and 0.89–0.93 at $n = 32$ for single-second perturbations, with cost roughly proportional to the number of admitted cells.

5 Joint score bound

After propagating all T steps, the residuals $r = \mathrm{AE}(x) - x \in \mathbb{R}^N$ are affine in the *same* symbol vector $\alpha \in [-1, 1]^P$ (input symbols and all fresh symbols):

$$\hat{r} = c + V\alpha, \qquad c \in \mathbb{R}^N,\ V \in \mathbb{R}^{N\times P}. \tag{14}$$

By (I) and (1), $\max_{x\in X_\epsilon} s(x) \le \frac{1}{N} \max_{\alpha\in[-1,1]^P} \|c + V\alpha\|_2^2$.

Componentwise bound (Cert-RNN).

$$\bar{s}_{\mathrm{comp}} = \frac{1}{N}\sum_{k=1}^{N} \big(|c_k| + \|V_{k,:}\|_1\big)^2. \tag{15}$$

Each term is the worst case of one residual, attained at its own α; the sum lets every residual use a different α, which cannot happen because there is only one.

Theorem 11 (Joint score bound)**.**

$$\max_{\alpha\in[-1,1]^P} \|c + V\alpha\|_2^2 \ \le\ \|c\|_2^2 + 2\left\|V^\top c\right\|_1 + \sum_{i,j=1}^{P} \left|(V^\top V)_{ij}\right| \ =:\ N\,\bar{s}_{\mathrm{joint}}, \tag{16}$$

and $\bar{s}_{\mathrm{joint}} \le \bar{s}_{\mathrm{comp}}$.

Proof. Expand $\|c + V\alpha\|_2^2 = \|c\|_2^2 + 2(V^\top c)^\top\alpha + \alpha^\top(V^\top V)\alpha$. With $|\alpha_i| \le 1$: $(V^\top c)^\top\alpha \le \sum_i \left|(V^\top c)_i\right|$ and $\alpha^\top M\alpha = \sum_{i,j} M_{ij}\alpha_i\alpha_j \le \sum_{i,j} |M_{ij}|$, which gives (16). For the comparison, write $(|c_k| + \|V_{k,:}\|_1)^2 = c_k^2 + 2\,|c_k| \sum_i |V_{ki}| + \sum_{i,j} |V_{ki}|\,|V_{kj}|$ and sum over k. Then $\sum_k c_k^2 = \|c\|_2^2$; $\sum_k |c_k|\,|V_{ki}| \ge |\sum_k c_k V_{ki}| = \left|(V^\top c)_i\right|$; and $\sum_k |V_{ki}|\,|V_{kj}| \ge |\sum_k V_{ki}V_{kj}| = \left|(V^\top V)_{ij}\right|$. Hence $N\bar{s}_{\mathrm{comp}} \ge N\bar{s}_{\mathrm{joint}}$. □

Remark 12. The inequality is strict whenever residuals *cancel*: for example, when one tampered reading pushes some residuals up and others down, the terms $c_k V_{ki}$ or $V_{ki}V_{kj}$ have opposite signs across k and the absolute values in (16) are taken *after* summing over k. The implementation returns $\min(\bar{s}_{\mathrm{joint}}, \bar{s}_{\mathrm{comp}})$, since both are sound. Computing (16) costs one $P \times P$ Gram product.

6 Comparison with box-based linear relaxations (CROWN)

CROWN [12] and its generalisation to arbitrary computational graphs, auto_LiRPA [13], also keep linear dependence on the input: the final bound is a linear function of x, obtained by back-substitution from the score. The difference lies in how each non-linear operation is relaxed. CROWN relaxes each activation using only that neuron's own pre-activation interval, and auto_LiRPA, which implements CROWN for general graphs and is the version used in our benchmark, relaxes a product $f(x, y)$ with linear lower and upper bounds valid on the *box* $[l_x, u_x] \times [l_y, u_y]$ given by each operand's own interval bounds, in the style of McCormick envelopes [11]. In the notation above this is $S = \mathcal{B}_{xy}$ (with CROWN's own interval bounds), which, by the lemma on the joint region, includes corners that no input can reach, and these are where the extrema of $f - Ax - By$ typically lie.

Two consequences, both visible in the benchmark:

- α-CROWN [14] optimises the *slopes* of the relaxations but keeps the box as the region of validity; it cannot remove the slack due to unreachable corners. On the main benchmark it verified 142 queries against CROWN's 141.
- Branch and bound (β-CROWN [15], with general non-linear branching GenBaB [16]) splits operand ranges into smaller boxes, each containing fewer unreachable combinations, at a large time cost (159 verified at a 1800 s per-query limit, 23 timeouts). Cert-RNN+ removes the unreachable region directly (Theorem 10) and bounds the score jointly (Theorem 11), verifying 172 of 182 verifiable queries with a median of 3.4 s per query.

Box-based relaxations do have a cost advantage: their per-operation cost does not grow with the number of noise symbols, whereas Cert-RNN+ carries every input symbol and every fresh symbol, so its cost grows with model width.

7 Summary of the soundness argument

1. The input box is represented exactly as a zonotope (one symbol per attacked reading).
2. Affine operations are exact; unary activations and bilinear products introduce one fresh symbol per coordinate whose coefficient bounds the linearisation error over a region that contains every reachable value. Invariant (I) therefore holds at every step.
3. For bilinear products, Cert-RNN+ uses the region S_+ with $Z_{xy} \subseteq S_+ \subseteq \mathcal{B}_{xy}$, which is sound and never looser than the box (Theorem 10).
4. The score is bounded by (16), which is sound and never looser than (15) (Theorem 11).
5. Hence $\max_{x \in X_\epsilon} s(x) \leq \min(\bar{s}_{\text{joint}}, \bar{s}_{\text{comp}})$, and a result $\leq \tau$ certifies the property.

Floating-point implementation details (outward rounding of the computed bounds) are documented separately in the project's soundness notes.

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